Sophia Tang

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EDUCATION

University of Pennsylvania Philadelphia, PA
Dual Degree with Bachelor of Science in Engineering and Bachelor of Science in Economics Sept 2023 - May 2027 (*expected*)
Jerome Fisher Program in Management & Technology; Major in Computer Science & Statistics
Relevant Coursework: Algorithms, Programming Languages and Techniques I and II, Mathematics of Computer Science, Multivariable Calculus, Linear Algebra, Machine Learning, Computer Systems, Automata Computability and Complexity, Corporate Finance, Database & Information Systems, Statistical Inference, Convex Optimization.

RESEARCH EXPERIENCE

Kempner Institute, Harvard University. Boston, MA
Visiting Undergraduate Researcher. **Advised by Dr. Michael Albergo**. May 2026 - Present
- Developing theoretical generative modeling frameworks.

Chatterjee Lab, University of Pennsylvania. Philadelphia, PA
Undergraduate Researcher. **Advised by Dr. Pranam Chatterjee**. Sept 2024 - Present
- Developing theoretical ML frameworks for generative modeling with applications in biological design and simulation.
- **First-author papers published at ICML 2025 and ICLR 2026, and several papers under review** (**see below**).

Mitchell Lab, University of Pennsylvania. Philadelphia, PA
Undergraduate Researcher. **Advised by Dr. Michael Mitchell.** Sept 2023 - Present
- Developed novel targeted lipid nanoparticle (LNP)-encapsulated mRNA therapeutics with high transport across the blood-brain barrier after systemic administration and transfection into brain cell types.
- **Several published co-authored papers** and a **first-author review paper (see below).**

MACHINE LEARNING PUBLICATIONS (*denotes equal contribution; † denotes advising role)

[1] PepTune: De Novo Generation of Therapeutic Peptides with Multi-Objective-Guided Discrete Diffusion.
Sophia Tang*, *Yinuo Zhang*, Pranam Chatterjee*†.
International Conference on Machine Learning (**ICML 2025**), Dec 2024. **https://openreview.net/forum?id=FQoy1Y1Hd8**
Description: We introduce Monte Carlo Tree Guidance (MCTG), an inference-time multi-objective guidance algorithm that balances exploration and exploitation to iteratively refine Pareto-optimal sequences. MCTG integrates classifier-based rewards with search-tree expansion, overcoming gradient estimation challenges and data sparsity.

[2] Branched Schrödinger Bridge Matching.
***Sophia Tang**, Yinuo Zhang, Alexander Tong†, Pranam Chatterjee†*.
International Conference on Learning Representations (**ICLR 2026**). Jun 2025. **https://doi.org/10.48550/arXiv.2506.09007**
Description: We introduce Branched Schrödinger Bridge Matching (BranchSBM), a novel framework that learns branched Schrödinger bridges. BranchSBM parameterizes multiple time-dependent velocity fields and growth processes, enabling the representation of population-level divergence into multiple terminal distributions.

[3] Foundations of Schrödinger Bridges for Generative Modeling
Sophia Tang
Preprint. Mar 2026. **https://doi.org/10.48550/arXiv.2603.18992**
Description: This book (220 pages, 24 figures) covers the mathematical foundation of Schrödinger bridges as a unifying principle underlying modern generative modeling techniques.

[4] TR2-D2: Tree Search Guided Trajectory-Aware Fine-Tuning for Discrete Diffusion.
Sophia Tang*, *Yuchen Zhu*, Molei Tao†, Pranam Chatterjee†*.
Preprint. Sept 2025. **https://doi.org/10.48550/arXiv.2509.25171**
Description: We introduce a novel framework that optimizes reward-guided discrete diffusion trajectories with tree search to construct replay buffers for trajectory-aware fine-tuning.

[5] Entangled Schrödinger Bridge Matching.
***Sophia Tang**, Yinuo Zhang, Pranam Chatterjee†*.
Preprint. Oct 2025. **https://doi.org/10.48550/arXiv.2511.07406**
Description: We introduce a framework that learns the first- and second-order stochastic dynamics of interacting, multi-particle systems where the direction and magnitude of each particle's path depend dynamically on the paths of the other particles.

[6] A2D2: Finetuning Any-Length Discrete Diffusion for Adaptive Decoding.
***Sophia Tang**, Yuchen Zhu, Molei Tao†, Pranam Chatterjee†*.
Preprint. June 2026. **https://doi.org/10.48550/arXiv.2606.13565**

[7] Gumbel-Softmax Flow Matching with Straight-Through Guidance for Controllable Biological Sequence Generation.
Sophia Tang, *Yinuo Zhang, Alexander Tong†, Pranam Chatterjee†.*
Preprint. Mar 2025. **https://doi.org/10.48550/arXiv.2503.17361**

[8] mRNAutilus: Multi-Objective-Guided Discrete Generation of mRNA with Optimized Therapeutic Properties.
Sawan Patel,* ***Sophia Tang****, *Yesol Kim*, Yinuo Zhang, Divya Srijay, Ping-Jung Lin, Shambhavi Shubham, Fengmei Pi, Cedric Wu, Sherwood Yao†, and Pranam Chatterjee†.*
Preprint. Jun 2026. **https://doi.org/10.48550/arXiv.2605.31296**

[9] PeptiVerse: A Unified Platform for Therapeutic Peptide Property Prediction.
Yinuo Zhang, ***Sophia Tang***, *Tong Chen , Elizabeth Mahood , Sophia Vincoff and Pranam Chatterjee†.*
Nature Communications (In Print). Jan 2026. **https://www.biorxiv.org/content/10.64898/2025.12.31.697180v1**

[10] TD3B: Transition-Directed Discrete Diffusion for Allosteric Binder Generation.
Hanqun Cao, Aastha Pal*,* ***Sophia Tang***, *Yinuo Zhang, Jingjie Zhang, Pheng Ann Heng, Pranam Chatterjee†*
International Conference on Machine Learning (**ICML 2026**). May 2026. **https://arxiv.org/abs/2605.09810**

[11] Multi-Objective-Guided Discrete Flow Matching for Controllable Biological Sequence Design.
Tong Chen, Yinuo Zhang, ***Sophia Tang***, *and Pranam Chatterjee†.*
Preprint. May 2025. **https://doi.org/10.48550/arXiv.2505.07086**

[12] Active Flow Expansion for Out-of-Distribution Discovery: from Theory to Molecules.
Riccardo De Santi, Bruce Lee, Cristian Perez Jensen, Kimon Protopapas, ***Sophia Tang***, *Cheng-Hao Liu, Pranam Chatterjee, Yisong Yue, Andreas Krause.*
Preprint. June 2026. **https://doi.org/10.48550/arXiv.2606.08802**

BIOENGINEERING PUBLICATIONS († denotes advising role)

[13] Peptide-functionalized nanoparticles for brain-targeted therapeutics.
Sophia Tang, *Emily L. Han, and Michael J. Mitchell†.*
Drug Delivery and Translational Research, Springer Nature. Mar 2025. **https://doi.org/10.1007/s13346-025-01840-w**

[14] Peptide-Functionalized Lipid Nanoparticles for Targeted Systemic mRNA Delivery to the Brain.
Emily L. Han, ***Sophia Tang***, *Dongyoon Kim, Amanda M. Murray, Kelsey L. Swingle, Alex G. Hamilton, Kaitlin Mrksich, Marshall S. Padilla, Rohan Palanki, Jacqueline J. Li, and Michael J. Mitchell†.*
ACS Nano Letters. Dec 2024. **https://doi.org/10.1021/acs.nanolett.4c05186**

[15] Predictive High-Throughput Platform for Dual Screening of mRNA Lipid Nanoparticle Blood–Brain Barrier Transfection and Crossing.
Emily L. Han, Marshall S. Padilla, Rohan Palanki, Dongyoon Kim, Kaitlin Mrksich, Jacqueline J. Li, ***Sophia Tang***, *Il-Chul Yoon, and Michael J. Mitchell†.*
ACS Nano Letters. Jan 2024. **https://doi.org/10.1021/acs.nanolett.3c03509**

[16] Optimized microfluidic formulation and organic excipients for improved lipid nanoparticle-mediated genome editing.
Rohan Palanki, Emily L. Han, Amanda M. Murray, Rohin Maganti, ***Sophia Tang***, *Kelsey L. Swingle, Dongyoon Kim, Hannah Yamagata, Hannah C. Safford, Kaitlin Mrksich, William H. Peranteau, and Michael J. Mitchell†.*
Lab on a Chip, Royal Society of Chemistry. Jan 2024. **https://doi.org/10.1039/D4LC00283K**

[17] High-Throughput In Vivo Screening Identifies Structural Factors Driving mRNA Lipid Nanoparticle Delivery to the Brain.
Emily L Han, Dongyoon Kim, Amanda M Murray, Kaitlin Mrksich, Alex G Hamilton, ***Sophia Tang***, *Soyeon Yoo, Audrey T Zhu, Elaine R Tong, Rohan Palanki, Michelle L Hall, Sean K Bedingfield, Michael J Mitchell†.*
ACS Nano. Jan 2026. **https://doi.org/10.1021/acsnano.5c19138**

TECHNICAL ARTICLES

A Complete Guide to Spherical Equivariant Graph Transformers. **https://doi.org/10.48550/arXiv.2512.13927**
A 99-page breakdown of the theory behind spherical equivariant (SE(3)) graph neural networks (EGNNs).

A Complete Guide to Protein Folding Prediction. **https://alchemybio.substack.com/p/a-complete-guide-to-protein-folding**
A 3-hour breakdown of the three-track RoseTTAFold protein prediction model that leverages multi-modal deep learning architectures.

INVITED TALKS

Discrete Diffusion Reading Group. Nov 2025. **https://youtu.be/NX2Mth_Epvk?si=LsVS_bhzyD-RV03m**

Starkly Speaking Reading Group. Jul 2025. **https://youtu.be/inVYA0pQ4Wg?si=mdwhTE3wu0vXJ4ZX**

Learning on Graphs and Geometry Reading Group. Jan 2025. **https://youtu.be/KVr8ryclwdA?si=RdRPdqNqvmblk3Og**

AWARDS AND HONOURS

CRA Outstanding Undergraduate Researcher Award. Jan 2026. 1 of 8 awardees across North America. The award is widely regarded as the highest honor for undergraduate computer science students and recognizes exceptional research accomplishments in computing.
https://www.seas.upenn.edu/stories/sophia-tang-wins-cra-outstanding-undergraduate-research-award/